Epigraphic Masked Language Models for Dead Sea Scrolls Restoration: Unifying Token Alignment, Physical Ink Traces, and Multi-Word Length Ensembling

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August 10, 2026

Abstract

Restoring missing text in 2,000-year-old Dead Sea Scrolls (lacunae) is a fundamental challenge in biblical studies and digital humanities. Drawing inspiration from recent pioneering applications of Masked Language Modeling (MLM) to cuneiform tablets ([4]), ancient Hebrew inscriptions ([3]), and medieval Hebrew manuscripts ([5]), we present a comprehensive epigraphic restoration framework specifically engineered for non-biblical Dead Sea Scrolls. We demonstrate that subword tokenization (WordPiece) suffers a 38.3% token alignment drop rate on ancient character-bounded gaps, whereas character-level MLMs (TavBERT FT) eliminate alignment drops entirely (0.0% drops), achieving 23.65% Hit@10 on synthetic cloze (scatter-30, $n = 729$). On real physical manuscript lacunae, conditioning on surviving ink traces ($P0$) and physical hole bounds ($L \pm 1$) lifts Top-10 restoration accuracy from 9.46% ($U0$ unconstrained) to 64.86% ($P0$). Incorporating our 4-point algorithmic roadmap (morpho-lemma normalization, quote-aware Pesher RAG, IDF term boosting, and epigraphic stroke filtering) elevates accuracy to 66.22% Top-1 / 83.78% Top-10 / 87.84% Top-20 on peer-reviewed Qumran-Digital scholar targets ($n = 74$), more than tripling initial 1950s human scholar editions (20.27% Top-1). Finally, for unknown-length multi-word lacunae ($L \in [3, 15]$), our length-ensemble beam search (LengthEnsembleCharMLMGenerator) breaks the 0.0% structural collapse wall of standard single-length MLMs, achieving 14.2% Hit@10 / 14.5% slot accuracy.

1. Introduction

The Dead Sea Scrolls—discovered in eleven caves near Qumran between 1947 and 1956—represent the most significant manuscript discovery of the twentieth century, containing over 25,000 fragments of ancient Hebrew and Aramaic texts dated between the 3rd century BCE and the 1st century CE. Over two millennia of physical deterioration, moisture, and parchment rot have left extensive holes (lacunae) across these scroll fragments.

1.1. Lineage and Related Work

Recent advances in neural natural language processing have formulated ancient text restoration as a Masked Language Modeling (MLM) task:

1. Akkadian Cuneiform Tablets: Lazar et al. (EMNLP 2021) formulated cuneiform sign restoration as an MLM task over Oracc transliterations using mBERT and greedy decoding, achieving high sign accuracy on formulaic royal inscriptions.
2. Biblical Hebrew and Aramaic Inscriptions: Fono et al. (Embible / EACL 2024) applied transformer ensembles to clean biblical texts, comparing character and word models under synthetic masking.
3. Hebrew Manuscript Subword Modeling: Shmidman et al. (MsBERT / ML4AL 2024) introduced MsBERT (a WordPiece subword model) trained on medieval Rabbinic Hebrew transcriptions from Dicta.

1.2. The 3 Unresolved Epigraphic Bottlenecks

While these foundational studies established the viability of MLM for ancient texts, three critical methodological bottlenecks remained unresolved for Dead Sea Scrolls epigraphy:

1. Subword Alignment Failure Wall: Subword tokenizers (WordPiece / MsBERT) generate modern subword tokens that cannot align to exact physical character gap lengths, causing 38.3% token drop rates on ancient Hebrew content words.
2. Unconstrained Context Failure Wall ($U0$): Evaluating models on context alone ($U0$) yields poor real-world lacuna restoration ($< 10\%$ Top-10) because ancient context is heavily damaged. Physical ink conditioning ($P0$) is essential.
3. Multi-Word Fixed-Slot Collapse: Standard single-length BERT MLMs collapse to 0.0% accuracy on multi-word lacunae because fixed input token slots cannot adapt output sequence lengths when exact character counts (L) are unknown.

2. Dataset Architecture & Safeguards

Our dataset is constructed from the ETCBC Text-Fabric 2.0 digital corpus of non-biblical Dead Sea Scrolls (etcbc/dss), spanning 732 non-biblical manuscripts (268,594 total word positions, 103,355 fully preserved words rec=0, 165,239 damaged word positions rec=1, and 27,814 physical lacuna sites).

2.1. Three Golden Data Protection Rules

To eliminate data leakage, we enforce three strict methodological safeguards:

1. Reconstruction Redaction (rec=1 Filter): Every sign flagged rec=1 in Text-Fabric is an editorial reconstruction and is 100% stripped from input and training data. The model trains ONLY on authentic 2,000-year-old surviving ink (rec=0).

2. SHA-1 Manuscript-Disjoint Partitioning: Scrolls are partitioned deterministically by SHA1(scroll_id) mod 100 into 531 train (72.5%) / 108 val (14.8%) / 93 test (12.7%) with 0 straddling scrolls.

3. Biblical Exclusion: Biblical Dead Sea Scrolls (e.g., 1QIsa^{a}) are 100% excluded from model training to prevent canonical text memorization.

3. Methodology & Champion Pipeline

3.1. Character-Level Masked Language Modeling

We fine-tune TavBERT (tau/tavbert-he), a character-level BERT model pretrained on Hebrew text, using 128-token continuous sliding windows over preserved non-biblical scroll text ($\geq$ 20 preserved words per chunk). Because 1 token = 1 character, TavBERT eliminates subword token alignment failures (0.0% drop rate).

3.2. Physical Ink & Hole Bounds Conditioning ($P0$)

During beam search generation, candidate words are filtered by PartialLetterFilter:

$$\mathrm{Len}(c) \in [L-1, L+1] \wedge \forall i \in \mathrm{SurvivingInk}(P0) : c[i] = P0[i] \tag{1}$$

This physical constraint eliminates 99% of dictionary candidates before candidate ranking.

3.3. The 4-Point Algorithmic Roadmap

To maximize real lacuna restoration accuracy, we construct a 4-point post-processing roadmap:

1. Step 1 (Morpho-Lemma Normalization): Normalizes inflectional variants (plene vs. defective spelling, gender/number suffixes).

2. Step 2 (Quote-Aware Pesher RAG): Detects citation markers (peshro ‘al) in commentary scrolls (*Pesharim*) via PesherQuoteRetriever, identifying the quoted biblical book (86.57% Top-1 book recovery) and injecting thematic context.

3. Step 3 (Sectarian IDF Boosting): Applies SectarianIDFBooster to down-weight ultra-generic Hebrew particles in favor of domain-specific sectarian vocabulary.

4. Step 4 (Epigraphic Stroke Filter): EpigraphicStrokeFilter permits stroke-ambiguous letter substitutions (e.g., resh vs. dalet vs. vav), reflecting physical ink erosion.

3.4. Multi-Word Length-Ensemble Beam Search

For multi-word lacunae of unknown character length ($L \in [3, 15]$), LengthEnsembleCharMLMGenerator loops character mask lengths $L = 3, 4, \ldots, 15$ in TavBERT and re-ranks all generated phrases across lengths using length-penalty probability scoring:

$$\mathrm{Score}(c) = \frac{\sum_{i=1}^{L} \log P(c_i)}{\sqrt{L}} \tag{2}$$

4. Experimental Results

4.1. RQ1: Synthetic Cloze Architecture Benchmark

We evaluate raw model architecture quality on 100 held-out test sentences ($n = 729$ masked content words, scatter-30, Track A) under gold character length (O-len).

Table 1: Synthetic Cloze Restoration Benchmark (scatter-30, $n = 729$). 95% CIs are sentence-level percentile cluster bootstrap ($B = 1,000$).

Model Variant	Hit@10	Hit@1	Drops
TavBERT FT (Optimal)	23.65%	8.42%	0.0%
DictaBERT-char FT	22.80%	8.10%	0.0%
TavBERT Base	21.12%	7.27%	0.0%
MsBERT Fine-Tuned	13.31%	9.78%	38.3%
ByT5 Fine-Tuned	5.35%	0.82%	0.0%

Key Takeaway (RQ1): Character-level MLMs (TavBERT FT 23.65%) decisively outperform subword WordPiece models (MsBERT 13.31%) by eliminating token alignment drops (38.3% → 0.0%).

4.2. RQ2: Real Lacuna Scholar Agreement Benchmark

We evaluate literature agreement against peer-reviewed published scholar proposals from the Göttingen Qumran-Digital (QD) database ($n = 74$ real physical lacuna targets).

Key Takeaway (RQ2): Physical ink conditioning ($P0$) lifts Top-10 accuracy from 9.46% ($U0$) to 64.86% ($P0$). Our full 4-Point Roadmap reaches 66.22% Top-1

Table 2: Real Lacuna Literature Agreement Benchmark ($n = 74$ QD targets).

System / Condition	Regime	Top-1	Top-10	Top-20
TavBERT FT + Roadmap	P0+Road	66.22%	83.78%	87.84%
TavBERT FT Baseline	P0, ±1	47.30%	64.86%	70.27%
DictaBERT-char FT	P0, ±1	44.59%	66.22%	72.97%
MsBERT FT Baseline	P0, ±1	40.50%	63.50%	67.60%
Human Scholar Control	DJD 1950s	20.27%	43.24%	44.60%
Pure Dictionary Regex	Regex	8.12%	34.50%	41.20%
MsBERT FT (Unconstrained)	U0	—	9.46%	—

/ 83.78% Top-10, more than tripling initial 1950s human scholar editions (20.27% Top-1).

4.3. RQ3: Large-Scale Physical Test Split Benchmark

We evaluate physical lacuna restoration across all 3,695 single-word test lacunae in the 93 held-out test scrolls from Text-Fabric.

Table 3: Large-Scale Text-Fabric Physical Lacuna Benchmark ($n = 3,695$ test lacunae across 93 held-out scrolls).

Model Variant	Regime	Top-1	Top-10
TavBERT FT + Roadmap	P0+Road	65.80%	82.90%
TavBERT FT Baseline	P0 Exact	46.80%	64.50%
DictaBERT-char FT	P0 Exact	44.20%	65.80%
MsBERT Fine-Tuned	P0 Exact	39.80%	63.10%
MsBERT FT (Unconstrained)	U0 Context	2.10%	9.20%

4.4. RQ4: Unknown-Length Multi-Word Lacuna Benchmark

We evaluate multi-word lacunae of unknown character length ($L \in [3, 15]$) using LengthEnsembleCharMLMGenerator.

Table 4: Unknown-Length Multi-Word Lacuna Benchmark.

Model Variant / Strategy	Gap Condition	Hit@10 / Acc.
LengthEnsembleCharMLM	Multi-Word Slot	14.5% (14.2%)
Fixed Single-Length MLM	Unknown $L \in [3, 15]$	0.0% (Collapse)
Fixed Length Oracle (O-len)	1-Word Known	16.7% (Upper Bound)

5. Comparative Synthesis Against Prior Art

Table 5 presents a direct comparative synthesis comparing our system against the three foundational reference papers.

6. Conclusion

By combining character-level Masked Language Models, physical ink trace filtering ($P0$), a 4-point algorithmic roadmap, and length-ensemble decoding, our framework provides a complete, leak-free solution for ancient manuscript text restoration. Our full system achieves 66.22% Top-1 / 83.78% Top-10 agreement with peer-reviewed Dead Sea Scrolls scholar proposals, more than tripling early 1950s human scholar editions (20.27% Top-1). All code, models, and evaluation benchmarks are released open-source.

References

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Table 5: Comparative Synthesis Against Prior Epigraphic Works.

Dimension	Lazar et al. (2021)	Fono et al. (2024)	Shmidman et al. (2024)	Our System
Target Language	Akkadian (Cuneiform)	Ancient Hebrew / Aramaic	Rabbinic Hebrew Manuscripts	Dead Sea Sc
Tokenizer Modality	Subword WordPiece	Character & Word Ensemble	Subword WordPiece (MsBERT)	Character M
Alignment Drops	Not quantified	Not quantified	38.3% Drop Rate	0.0% Drop R
Physical Ink Traces	None (Sign count 'x')	None	None	P0 Ink Filte
Multi-Word Decoding	Greedy token decoding	Fixed length masking	Fixed length masking	LengthEnsen
Scholar Control	Blind 3-expert rating	None	None	70-Year QD
Split Safeguards	Random 80/20 train/test	Random 80/20 train/test	Random train/test	SHA-1 Disjo
Top Score	89.5% Hit@5 (Sign)	53.2% Hit@10 (Biblical)	Not reported on physical DSS	83.78% Top-